Exercise J-M

1. Let PQR be triangle. The point A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR}=\frac{RB}{BP}=\frac{PC}{CQ}=\frac{1}{2}$. Then $\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)}$ is equal to : **[JEE(Main)Jan.2023]**

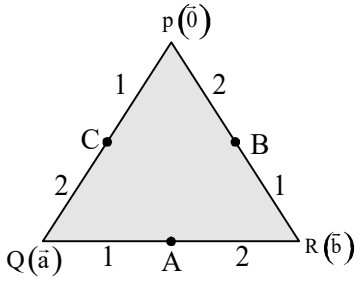
माना PQR एक त्रिभुज है। भुजाओं QR, RP तथा PQ पर क्रमशः बिंदु A, B तथा C इस प्रकार है कि $\frac{QA}{AR}=\frac{RB}{BP}=\frac{PC}{CQ}=\frac{1}{2}$ है,

तो $\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 2 (B) 3 (C) 4 (D) $\frac{5}{2}$

Ans. B

Sol.



Let position vector of P, Q, R be $\bar{0}, \bar{a}$ and $\bar{b}$ respectively

$\Rightarrow$ P.V. of $C=\frac{\bar{a}}{3}$, P.V of $A=\frac{2\bar{a}+\bar{b}}{3}$ and P.V of $B=\frac{2\bar{b}}{3}$

$\therefore \overline{CA}=\frac{\vec{a}+\vec{b}}{3}$ and $\overline{AB}=\frac{\bar{b}-2\bar{a}}{3}$

$\Delta PQR=\frac{1}{2}|\overline{PQ}\times\overline{PR}|=\frac{1}{2}|\bar{a}\times\bar{b}|$

$\Delta ABC=\frac{1}{2}|\overline{CA}\times\overline{AB}|=\frac{1}{2}\left|\left(\frac{(\bar{a}+\bar{b})}{3}\times\frac{(\bar{b}-2\bar{a})}{3}\right)\right|=\frac{1}{2}\left|\frac{\bar{a}\times\bar{b}}{3}\right|$

$=\frac{\triangle PQR}{\triangle ABC}=3$

2. For a triangle ABC, the value of $\cos 2A+\cos 2B+\cos 2C$ is least. If its inradius is 3 and incentre is M, then which of the following is NOT correct? **[JEE(Main)Jan.2023]**

(A) $\overrightarrow{MA}\cdot\overrightarrow{MB}=-18$

(B) perimeter of ΔABC is $18\sqrt{3}$

(C) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(D) area of ΔABC is $\frac{27\sqrt{3}}{2}$

एक त्रिभुज ABC के लिए $\cos 2A + \cos 2B + \cos 2C$ का मान न्यूनतम है। यदि इसके अंतःवृत्त की त्रिज्या 3 तथा अन्तःवृत्त का केन्द्र M है, तब निम्न में से कौन सा सही नहीं है ? **[JEE(Main)Jan.2023]**

(A) $\overrightarrow{MA} \cdot \overrightarrow{MB} = -18$

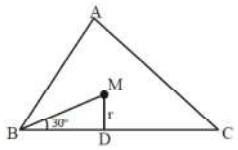
(B) ΔABC का परिमाण $18\sqrt{3}$

(C) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(D) ΔABC का क्षेत्रफल $\frac{27\sqrt{3}}{2}$

Ans. D

Sol.



If $\cos 2A + \cos 2B + \cos 2C$ is minimun then $A = B = C = 60^o$

So ΔABC is equilateral

Now in–radias $r = 3$

So in ΔMBD we have

$$\tan 30^o = \frac{MD}{BD} = \frac{r}{a/2} = \frac{6}{a}$$

$$\frac{1}{\sqrt{3}} = \frac{1}{a} = a = 6\sqrt{3}$$

Perimeter of $\Delta ABC = 18\sqrt{3}$

Area of $\Delta ABC = \frac{\sqrt{3}}{4}a^2 = 27\sqrt{3}$

3. In a triangle ABC, if $\cos A + 2\cos B + \cos C = 2$ and the lengths of the sides opposite to the angles A and C are 3 and 7 respectively, then $\cos A - \cos C$ is equal to : **[JEE(Main)April,2023]**

यदि एक त्रिभुज ABC में $\cos A + 2\cos B + \cos C = 2$ है तथा कोणों A तथा C के सम्मुख भुजाओं की लंबाई क्रमशः 3 तथा 7 है, तो $\cos A - \cos C$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{9}{7}$ (B) $\frac{5}{7}$ (C) $\frac{3}{7}$ (D) $\frac{10}{7}$

Ans. D

Sol. $\cos A + \cos C = 2(1 - \cos B)$

$$2\cos\frac{A+C}{2}\cos\frac{A-C}{2}=4\sin^2\frac{B}{2}$$

as $\cos\left(\frac{A+C}{2}\right)=\sin\frac{B}{2}$

so $\cos\frac{A-C}{2}=2\sin\frac{B}{2}$

$$2\cos\frac{B}{2}\cos\frac{A-C}{2}=4\sin\frac{B}{2}\cos\frac{B}{2}$$

$$2\sin\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)=4\sin\frac{B}{2}\cos\frac{B}{2}$$

$\sin A+\sin C=2\sin B$

$a+c=2b\Rightarrow a=3,c=7,b=5$

$$\cos A-\cos C=\frac{b^2+c^2-a^2}{2bc}-\frac{a^2+b^2-c^2}{2ab}$$

$$=\frac{25+49-9}{70}-\frac{9+25-49}{30}$$

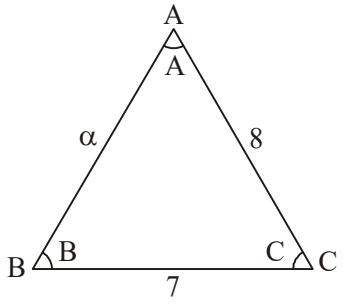
$$=\frac{65}{70}+\frac{1}{2}=\frac{20}{14}=\frac{10}{7}$$

4. In a triangle ABC, BC = 7, AC = 8, AB = $\alpha \in N$ and $\cos A=\frac{2}{3}$. If 49 cos(3C) + 42 = $\frac{m}{n}$, where gcd(m, n) = 1, then m + n is equal to _____. **[JEE(Main)April,2024]**

एक त्रिभुज ABC में, BC = 7, AC = 8, AB = $\alpha \in N$ तथा $\cos A=\frac{2}{3}$ हैं। यदि 49 cos(3C) + 42 = $\frac{m}{n}$, जहाँ gcd(m, n) = 1, तो m + n बराबर है _____ **[JEE(Main)April,2024]**

Ans. 39

Sol.



$$\cos A=\frac{2}{3}=\frac{\alpha^2+8^2-7^2}{2.\alpha.8}$$

$\Rightarrow (\alpha^2 + 15)3 = 32\alpha$

$3\alpha^2 - 32\alpha + 45 = 0$

$\Rightarrow \alpha = \frac{5}{3}, 9$

$\alpha \in N \Rightarrow \alpha = 9$

$\cos C = \frac{7^2 + 8^2 - 9^2}{2.7.8} = \frac{2}{7}$

$\cos 3C = 4\cos^3 C - 3 \cos C$

$= \frac{4 \times 8}{7^3} - \frac{6}{7}$

$49 \cos 3C = \frac{32}{7} - 42$

$\Rightarrow 49 \cos 3C + 42 = \frac{32}{7}$

$\Rightarrow m + n = 39$